

EX. NO: 3

IMPLEMENTATION OF HILL CIPHER

AIM

To write a C program to implement the Hill Cipher substitution technique for encryption using a 2×2 key matrix.

ALGORITHM

Step 1: Start the program.

Step 2: Read the plaintext from the user.

Step 3: Read the elements of the 2×2 key matrix.

Step 4: Convert each plaintext character into its numerical equivalent ($A = 0, B = 1, \dots, Z = 25$).

Step 5: Divide the plaintext into pairs of characters. If the length is odd, append the letter X.

Step 6: Encrypt each pair using the Hill Cipher formula:

$$C = (K \times P) \text{ } 26$$

where:

- K = Key Matrix
- P = Plaintext Vector
- C = Ciphertext Vector

Step 7: Convert the encrypted numerical values back to their corresponding alphabetic characters.

Step 8: Display the ciphertext.

Step 9: Stop the program.

PROGRAM:

```
#include <stdio.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
int main()
```

```
{
```

```
    int key[2][2];
```

```
    char text[100], cipher[100];
```

```
    int i, len;
```

```
    printf("Enter Plain Text: ");
```

```
    scanf("%s", text);
```

```

printf("Enter 2x2 Key Matrix (4 integers):\n");
for(i = 0; i < 2; i++)
    for(int j = 0; j < 2; j++)
        scanf("%d", &key[i][j]);

len = strlen(text);

// Convert to uppercase
for(i = 0; i < len; i++)
    text[i] = toupper(text[i]);

// Add X if length is odd
if(len % 2 != 0)
{
    text[len] = 'X';
    text[len + 1] = '\0';
    len++;
}

for(i = 0; i < len; i += 2)
{
    int p1 = text[i] - 'A';
    int p2 = text[i + 1] - 'A';

    cipher[i] = ((key[0][0] * p1 + key[0][1] * p2) % 26) + 'A';
    cipher[i + 1] = ((key[1][0] * p1 + key[1][1] * p2) % 26) + 'A';
}

cipher[len] = '\0';

printf("\nCipher Text: %s\n", cipher);

```

```
return 0;

}
```

OUTPUT:

The screenshot displays the Programiz C Online Compiler interface. The browser's address bar shows the URL `programiz.com/c-programming/online-compiler/`. The compiler window is titled "main.c" and contains the following C code:

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4
5 int main()
6 {
7     int key[2][2];
8     char text[100], cipher[100];
9     int i, len;
10
11     printf("Enter Plain Text: ");
12     scanf("%s", text);
13
14     printf("Enter 2x2 Key Matrix (4 integers):\n");
15     for(i = 0; i < 2; i++)
16         for(int j = 0; j < 2; j++)
17             scanf("%d", &key[i][j]);
18
19     len = strlen(text);
20
21     // Convert to uppercase
22     for(i = 0; i < len; i++)
23         text[i] = toupper(text[i]);
24 }
```

The "Output" panel on the right shows the execution results:

```
Enter Plain Text: pals
Enter 2x2 Key Matrix (4 integers):
3 3
2 5

Cipher Text: TEJI

=== Code Execution Successful ===
```

The bottom of the image shows a Windows taskbar with the system clock indicating 08:58 PM on 03-08-2026, and the weather showing 28°C with light rain.